

Load-Bearing but Redundant: Rethinking Physics Injection in Shared-Latent World Models

Anonymous submission

Abstract

A world model predicts how an environment evolves in response to actions by rolling an internal representation forward, and is expected to respect physical regularities such as how objects move and collide. We study JEPA-style latent world models and find a puzzle: trained with no physics supervision, such a model already recovers object position well from its representation, predicts one step almost exactly, and—as counterfactual intervention confirms—actually uses that state when rolling forward, yet its multi-step rollouts still drift away from the very dynamics it encodes. The popular fix is to inject physical structure into the latent representation. We test it systematically within the family it names—ten variants that write physics onto the coordinates of the shared latent, spanning hard and soft constraints on both physical quantities and physical laws, with labeled and label-free supervision—across three controlled physics domains and two realistic simulation benchmarks, and find injection fails in 29 of 30 cases; even supplying the *correct* physical constraint does not help. One account fits every test we ran: the physical state is already encoded, redundantly, across a *shared latent*—one representation the predictor runs through end-to-end—so an injected copy carries no new signal, while the constraint still competes for capacity and gradient. Intervening on a trained model shows the injected slot is *load-bearing but redundant*: perturbing it moves the model’s own black-box state $1.5\text{--}2.0\times$, and swapping it for another trajectory’s makes prediction follow that trajectory 47–72% of the way—yet the black box keeps its own copy of the same position, and prediction error does not move. Both routes carry load; the second one carries nothing new. What the model lacks is not the object’s physical state but the ability to evolve that state over long horizons: fixing the training procedure alone, adding no physics, cuts rollout error $2.2\text{--}8.3\times$ on identical code and metrics. To help a latent world model, injected structure should supply what the representation is missing—*how the state evolves, not what the state is*.

1 Introduction

A *world model* rolls an internal representation forward to predict how an environment evolves under an agent’s actions—a simulator to plan with instead of acting in the world (Ha and Schmidhuber 2018; LeCun 2022). For a physical environment, it should respect how objects move, fall, and collide. JEPA-style latent world models (Maes 2025; Balestrieri and LeCun 2025) present a puzzle here.

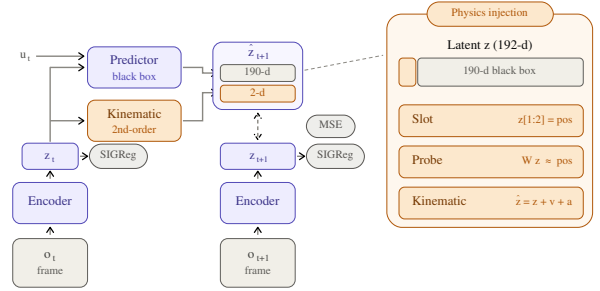


Figure 1: Physics injection into the LeWM shared-latent world model. *Amber* marks everything injection adds to the backbone; the rest is unmodified LeWM. The encoder maps each frame to a 192-d latent; a black-box transformer predicts the full latent from the latent history and action u_t , and the free-rollout objective aligns the predicted $\hat{z}_{t+1}$ to the encoded next latent by MSE (SIGReg prevents collapse). Physics enters a 2-d position *slot* carved from the 192 dimensions (right panel): a hard *slot* pinning $z[1:2]$ to position, a supervised *probe* reading position off the latent, or a *kinematic* head evolving the slot by a second-order rule—while the other 190 “black-box” dimensions stay free. This 2-vs-190 split is the crux: the black box can carry the same physical state, leaving prediction a route that need never pass through an injected slot.

Trained on PhyWorld (Kang et al. 2025) with no physics supervision, such a model already recovers object position from its representation by linear probing (correlation up to 0.96) and predicts one step almost exactly (latent cosine 0.98–0.99)—yet rolled out autoregressively, its predictions drift away from the very dynamics it encodes: on collision the latent cosine falls to 0.24 by horizon 28, and out-of-distribution (OOD) physics—unseen radii, masses, or velocities—multiplies rollout error several-fold. This indicates the state is already present in the representation; the model simply fails to *evolve* it.

The popular fix is to *inject physical structure into the latent* (Figure 1): a dedicated slot pinned to the physical state,

a supervised probe that reads it out, or a hard-wired dynamics rule. Such injection has real successes in adjacent model classes—architectures that route prediction through a low-dimensional physical state by construction (Mao, Umasudhan, and Ruchkin 2024; Peper et al. 2025), deep supervision where the supervised quantities occupy a large fraction of a compact state latent (Zahorodnii 2025), and a decomposed JEPA latent supervised on low-dimensional numerical time series (Nie et al. 2026). Scaling data is not the answer either: video generators trained at scale still fail OOD physics, memorizing cases rather than laws (Kang et al. 2025). But for a *shared, high-dimensional* latent—where prediction runs end-to-end through one representation—injection has only been tried in isolated, incomparable settings, and never against the confound that dominates the outcome: the training protocol. Whether physical structure adds anything, once the protocol is done right, has never been tested systematically.

We answer this with a systematic, controlled anatomy on a JEPA world model. We cross ten injection variants—spanning hard and soft constraints, targeting the physical *state* (what it is) or its *evolution* (how it changes), under labeled and label-free supervision—with two training regimes and three controlled physics domains, then corroborate on two realistic-simulation benchmarks (Bear et al. 2021; Tung et al. 2023).

Across the full grid, the structure does not help: injection degrades or fails to improve prediction in 29 of 30 mechanism-domain cells. Supplying the *correct* evolution rule—the gravitational law $a=g$ wired into the state update—does no better than a freely learned acceleration where $a=g$ is the ground truth (0.151 vs. 0.149 nMSE on parabola), and degrades uniform motion by $1.27\times$. Co-training the structure from scratch, rather than grafting it onto a pretrained encoder, does not rescue it either: over three seeds it remains harmful on uniform (+0.435 nMSE, against +0.037 fine-tuned) and on collision, and reaches only parity on parabola—never help—closing the defense that physics must be baked in during pretraining. The one cell that does improve reverses sign on the other two domains: using it would mean knowing each domain’s dynamics in advance.

Why should the *correct* physics hurt? Because the physical state is *already there*. A pinned slot occupies 2 of the 192 latent dimensions, but the other 190 redundantly encode the same position—decoding it nearly as accurately as the full latent—so an injected copy adds no information the representation lacks, while the constraint still competes with prediction for capacity and gradient. Redundancy accounts for the absent gain, competition for the harm. What injection adds is not a route the prediction lacked but a *second* route to a state it already had, and intervening on a trained model shows prediction takes both: steering the slot moves the model’s own black-box state by $1.5\text{--}2.0\times$ while a norm-matched random direction moves it ≈ 0 , and swapping the slot for another trajectory’s makes prediction follow that trajectory 47–72% of the way where a model without a slot follows 1.5–11%. The injected slot is thus *load-bearing but redundant*—used, and carrying nothing the black box was

not already carrying. This also explains why strengthening it cannot help: if the slot’s $\sim 1\%$ share of the prediction loss were what starves it, up-weighting should remove the harm, and it does not—re-weighting to saturation recovers only *parity*, never a net gain, because the constraint was never short of gradient, only short of new information.

What the representation lacks is not the state but the ability to *evolve* it over long horizons, and that gap is fixable with no physics at all. Repairing the training protocol alone does it: free rollout, with the horizon matched to each domain’s dynamics, cuts rollout error $2.2\text{--}8.3\times$ on the identical code, metrics, and seeds that judged every injection cell. That pins the failure on the injected structure rather than on the pipeline or the yardstick. The lesson is constructive: an inductive bias must supply what a shared latent is missing (how the state evolves), not re-encode what it already holds. If that reading is right, only an architecture can close the route; no loss weight can. Porting one that makes the physical state the *sole* pathway for prediction (Mao, Umasudhan, and Ruchkin 2024) does close it, but it collapses once an OOD shift reaches its encoder (ρ 0.33 vs. 0.89): closing the bypass is necessary, but not sufficient. Every finding above replicates on a second, frozen-DINOv2 backbone.

This paper contributes:

- to our knowledge, the first systematic anatomy of physics injection in *shared-latent* visual world models—ten variants $\times$ two training regimes $\times$ three domains, plus two realistic-simulation benchmarks—finding injection no better than baseline in 29 of 30 cells, and replicating on a frozen-DINOv2 backbone (§4.2, §4.4);
- a mechanistic account, *load-bearing but redundant*, established by intervention rather than inferred: three independent interventions on trained models—steering, counterfactual patching, and per-dimension Jacobians—agree that the injected slot *is* used, while the 190 black-box dimensions decode position as accurately as all 192 and keep doing so once the slot is pinned. Prediction reads both copies of one state, so re-weighting the slot to saturation buys only parity. A faithful extrinsic port supports this from the outside: making the physical state the sole route does help, yet collapses under encoder-OOD shift (§4.3, §4.4);
- a controlled factorial showing the dominant variable is the training protocol, not physical structure: free rollout alone cuts rollout error $2.2\text{--}8.3\times$ with no physics added, doubling as the positive control against implementation and metric artifacts (§4.3).

2 Related Work

Physically structured world models. A growing line argues that world models should carry explicit physical state: physically interpretable models route prediction through a dedicated low-dimensional state (Mao, Umasudhan, and Ruchkin 2024; Peper et al. 2025); deep supervision adds physical readouts to the loss and helps in a low-dimensional state latent, where the supervised quantities occupy $\sim 38\%$ of the representation (Zahorodnii 2025); closest to ours, Phys-JEPA supervises a decom-

posed JEPA latent and reports gains (Nie et al. 2026)—but on numerical time series, not the visual latents we study, a distinction that reverses the conclusion (§4.3). Physics-informed losses (Raissi, Perdikaris, and Karniadakis 2019) and conservation-structured dynamics (Greydanus, Dzamba, and Yosinski 2019; Cranmer et al. 2020) constrain the model class; graph simulators over *explicit* particle state predict physics well (Battaglia et al. 2016; Sanchez-Gonzalez et al. 2020); architectural biases such as restricted attention let transformers recover force laws from numerical trajectories (Liu et al. 2026)—structure works when the physical state *is* the representation, not a small part of a shared one. We do not dispute these results in their original settings; we test the transferable core of these proposals inside a shared-latent *visual* JEPA model and find all of them unhelpful or harmful (§4.2); our extrinsic port (§4.4) attributes the gap to architecture, not to the physics.

Latent world models and physics benchmarks. JEPA-style models predict in representation space (Assran et al. 2023; Bardes et al. 2024; Balestrierio and LeCun 2025; Maes 2025); object-level masking structures the JEPA objective itself rather than the latent’s physical content (Nam et al. 2026); DINO-WM plans over frozen features (Zhou et al. 2025; Oquab et al. 2024), the design we adopt as our cross-backbone control; Dreamer-line models train on imagined rollouts (Ha and Schmidhuber 2018; Hafner et al. 2025). PhyWorld (Kang et al. 2025) supplies controlled single-mechanism physics videos with explicit in-distribution ranges, showing that scaled video generators memorize cases rather than laws; Physion and Physion++ (Bear et al. 2021; Tung et al. 2023) add realistic—not camera-captured—simulation. We apply PhyWorld’s OOD protocol to *latent* models rather than pixel-space generators.

Rollout training and exposure bias. Training predictors on their own outputs is a classic remedy for compounding error (Bengio et al. 2015; Venkatraman, Hebert, and Bagnell 2015; Goyal et al. 2016). We claim no novelty for free rollout; its role here is methodological (§3.4): the controlled factorial variable prior injection studies never controlled, and a positive control separating “structure does not help” from “the implementation or metrics are broken.”

3 Method: A Design Space of Physical Injection

We answer the question of §1 by construction: design the ways physics can be injected into a shared latent, state what each design is supposed to achieve, and lay out the experiments that test the mechanism behind it.

3.1 Setting and Notation

We study LeWM (Maes 2025), a compact JEPA-style action-conditioned world model (~ 10 M parameters): an encoder E_θ (ViT-tiny) maps frame o_t to a latent $z_t \in \mathbb{R}^D$ with $D=192$, and a causal-transformer predictor F_ϕ rolls latents forward from the history and action u_t , i.e. $\hat{z}_{t+1} = F_\phi(z_{\leq t}, u_t)$. Training minimizes a prediction loss, an anti-

collapse regularizer (Balestrierio and LeCun 2025), and—when physics is injected—an auxiliary physical loss:

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \beta \text{SIGReg}(z) + \lambda \mathcal{L}_{\text{phys}}. \quad (1)$$

Here $\mathcal{L}_{\text{pred}} = \frac{1}{D} \sum_i w_i (\hat{z}_{t+1,i} - z_{t+1,i})^2$ is a per-dimension weighted prediction error. By default $w_i=1$; *re-weighting* raises w_i to w on the 2 slot dimensions (w swept $1 \rightarrow 300$ in §3.3), while λ scales the soft auxiliary losses (probe, consistency, label-free priors). Throughout, w denotes the slot’s prediction-loss weight and λ an auxiliary-loss weight—two distinct dials. *Teacher forcing* trains F_ϕ on one-step transitions from encoded ground-truth context; *free rollout* feeds predictions back for H autoregressive steps ($H=8$ unless stated) and supervises the whole rollout. The kinematic head (Group 2 below) replaces the slot’s transition by a second-order update

$$\hat{z}_{t+1}^{[0:2]} = z_t^{[0:2]} + v_t \Delta t + \frac{1}{2} a \Delta t^2, \quad (2)$$

where v_t is the finite-difference slot velocity and the acceleration a is either a learned MLP(z_t, u_t) or a single learnable constant—the strict $a=g$ head. Unless stated otherwise the encoder is initialized from the public PushT checkpoint; §4.2 additionally studies from-scratch training.

3.2 Injection Design Space: Three Groups, Ten Variants

We study one family: *intrinsic coordinate injection*—physics written onto the coordinates of a latent the predictor already shares, by a loss or a hard assignment, leaving the encoder and the transition operator otherwise untouched. This is the family the “just add a physics term” proposal names, and within it we vary three orthogonal axes: injecting *what the motion state is* vs. *how it evolves*, under *hard* vs. *soft* constraints, with *labeled* vs. *label-free* supervision. Table 1 groups the ten variants by the first axis and gives the design rationale of each.

Two large families lie outside it, and our results do not speak to them. One changes what the representation *is*—object-centric slots and relational graphs (Locatello et al. 2020; Battaglia et al. 2016; Sanchez-Gonzalez et al. 2020), or equivariant features. The other changes the form of the *transition operator*—Hamiltonian, Lagrangian or symplectic integrators (Cranmer et al. 2020), neural ODEs, and learned contact or impulse operators. Both alter the architecture rather than annotate its coordinates, which §4.3 argues is where the leverage lies; we test one such design as an external control (§3.5) but do not survey the space.

Orthogonally, we cross injection with the *training regime*: fine-tuning a pretrained encoder vs. from-scratch co-training; with injection on/off this gives a 2×2 per domain. The design targets one specific objection. If injection is only ever added to an encoder that has already learned a black-box representation, the constraint arrives after the representation is settled and never gets to shape it—which is precisely what extrinsic architectures do differently, binding the physical state from initialization. Under that reading our negative result would say nothing about physics injection as such, only about grafting it late. Co-training everything from

Variant	What it does, and what it tests
<i>Group 1: inject the state</i> — what the motion is (position, velocity)	
slot (structpos)	<i>hard.</i> pin dims [0:2] to the true position (the object’s x, y coordinates; a fixed index range, not an attention slot (Locatello et al. 2020)): the most direct state injection
+reweight	<i>hard.</i> raise the slot’s share of the prediction loss ($w=30$, Eq. 1): if failure is due to the slot’s $\sim 1\%$ gradient share, this should rescue it
+velocity	<i>hard.</i> encode velocity too: tests whether it matters <i>which</i> quantity is injected, not just that something is
probe	<i>soft.</i> linear head reads position from the latent: replicates the deep-supervision recipe (Zahorodnii 2025) in a high-dimensional visual latent
+slot	<i>both.</i> soft + hard combination: tests complementarity
<i>Group 2: inject the evolution</i> — how it changes (a dynamics rule)	
free MLP	<i>hard.</i> attach $z+v+a$ update with learned a : upgrade from pinning state to pinning evolution
strict $a=g$	<i>hard.</i> exact gravitational form, one learnable constant: closes the “your physics was wrong” objection
consistency	<i>soft.</i> finite-difference velocity of the rolled-out positions must match ground truth; assumes no form for a , designed for collision impulses
<i>Group 3: drop the labels</i> — is ground truth <i>needed</i> ? (plus a control)	
label-free	<i>soft.</i> slot must evolve as smooth second-order dynamics, no ground truth: the only injection available for raw video
grounded	<i>soft.</i> same structure + true labels: isolates the label variable

Table 1: Ten injection variants in three groups. The groups follow the **target** axis—inject the *state* (what the motion is), its *evolution* (how it changes), or, dropping **supervision**, a label-free prior over that evolution with its labeled control. The **constraint** axis is marked per variant: *hard* forces a latent dimension to equal the physical quantity, *soft* adds an auxiliary loss that only encourages it. The axes are orthogonal within intrinsic coordinate injection; they do not reach designs that restructure the representation or the transition operator. All variants use the strongest training protocol (§3.4).

random initialization removes the asymmetry: the constraint is present from the first gradient step. For the objection to hold, injection would have to *help* in that regime, or at least stop hurting; both outcomes are visible in the Δ column of §4.2.

3.3 Mechanistic Hypothesis and Its Tests

Two properties are easy to conflate, and separating them is what the tests below are for. A channel is **load-bearing** if perturbing it changes the prediction—a causal property,

measurable only by intervening on a trained model. A channel is **marginal** if it supplies information the rest of the latent does not—an informational property, visible in whether adding it improves prediction. Probing establishes neither: recoverable information need not be information the model uses (Elazar et al. 2021). The two come apart exactly when a representation is redundant, which is the case we are in.

Hypothesis (the redundancy problem). The physical slot occupies 2 of 192 dimensions, while the black-box channel *redundantly* encodes the same position—so an injected slot duplicates what the latent already carries. Prediction may well read the injected copy (§4.3 shows it does), but reading a duplicate cannot supply information the representation lacks: injection adds no new signal while still consuming capacity and gradient share. We test this account in three ways, each admitting a specific outcome that would falsify it:

1. **Is the physical constraint injected?** Split the latent into the 2 slot dimensions and the 190 black-box dimensions and decode position from each separately, with a random-2-dimension control (Hewitt and Liang 2019). If the black box alone cannot decode position, the redundancy claim is false.
2. **Is the constraint acting at all?** Two measurements: how large the weighted physical auxiliary loss is relative to the prediction loss (a proxy for gradient magnitude), and whether the encoder’s effective dimensionality (participation ratio) changes. If neither moves, the constraint never took effect and the failure is a bug. If the auxiliary term dominates and dimensionality collapses, the constraint is acting—and competing with prediction.
3. **Does re-weighting the physical loss help?** Sweep the slot weight w of Eq. 1 from 1 to 300. If harm does not respond systematically to the weight, the gradient-share mechanism is wrong; if it responds but weighting to saturation still cannot rescue every domain, demonstrating that insufficient physical loss is not the root cause.
4. **Is the injected slot load-bearing?** The first three tests establish that a redundant route exists, that the constraint acts, and that its strength is not the limit—but none of them contains the variable that matters here, namely which channel prediction reads. We therefore intervene on trained models three ways: *steering* (offset the position one channel decodes and read the change off the *other* channel, against a norm-matched random direction), *counterfactual patching* (replace a channel with a donor trajectory’s values and measure how far the rollout follows the donor, against a model with no slot to replace), and *per-dimension Jacobians* (differentiate the predicted position with respect to each latent coordinate, no intervention designed at all). If the slot bears none of the load, all three should read zero on it.

3.4 The Training-Protocol Control: Free Rollout

Free rollout serves two methodological roles; neither is a novelty claim (it is scheduled sampling, Bengio et al. 2015).

Controlled factorial variable. Physical structure has never been compared against the training protocol on the same baseline; prior studies treat the protocol as background. We

set the protocol first—replacing teacher forcing with free rollout (§3.1)—and then ask whether structure adds incremental value. Teacher forcing hides compounding error at training time (*exposure bias*): the model never sees its own mistakes, then must consume them at deployment. Free rollout exposes them during training, injecting no physics whatsoever. We read the resulting gain as more than robustness to compounding error: supervising an H -step autoregressive rollout turns training itself into *long-horizon dynamics simulation*. Under teacher forcing the predictor is graded only on reproducing local one-step transitions; under free rollout it is graded on *evolving* the state over the full horizon, so the gradient rewards a self-consistent multi-step dynamics rather than a one-step map—precisely the capability the latent is missing (§1). On this reading, free rollout is an evolution-targeted training signal that requires no physics labels: it trains *how the state evolves* without touching *what the state is*. A second protocol variable, the rollout horizon H , must *match dynamics complexity*: domains with late causal events (collision; realistic-simulation scenes) need horizons long enough to contain them.

Positive control. A negative result invites two deflations—“your implementation is broken” or “your metrics cannot detect improvement.” A protocol change that produces large gains on identical code and identical metrics rules out both at once, pinning the failure on the injected structure itself.

3.5 External and Cross-Backbone Controls

Two controls probe our account from outside the LeWM backbone: one tests whether an architecture that *closes* the bypass succeeds where injection fails, the other whether the conclusions survive a different encoder.

Extrinsic port. We faithfully port the physically interpretable world model (PIWM) of Mao, Umasudhan, and Ruchkin (2024)—a three-stage pipeline (VAE encoder, 128-D; a supervised extractor to a low-dimensional physical state; known-equation dynamics with learnable constants)—onto the same PhyWorld domains and OOD protocol. The port tests our mechanism’s architectural prediction from the outside: extrinsic designs make the physical state the sole pathway prediction must route through, closing the bypass by construction.

Cross-backbone control. To test that the conclusions are not artifacts of one encoder, we replicate the core protocol on a second JEPA-family instantiation in the style of DINO-WM (Zhou et al. 2025): a *frozen* DINOv2-small encoder (Oquab et al. 2024)—generic self-supervised features that have never seen PhyWorld—with the trainable projector serving as adapter and the identical predictor stack, losses, seeds, and evaluation. The variant differs from LeWM in backbone provenance, dimensionality, and, crucially, in whether the injected losses can reshape the encoder at all.

4 Experiments

4.1 Setup

Model. The compact, fully controlled backbone of §3.1 (LeWM: E_θ , an MLP projector, an action encoder, and a 6-layer causal-transformer F_ϕ) allows the complete factorial

anatomy (~ 200 training runs), with headline configurations repeated over three seeds (3072/1234/42); remaining cells are single-seed and marked as such.

Datasets and OOD protocol. All benchmarks are *visual* world-modeling tasks (RGB video frames), not the numerical/tabular sequences of prior physics-injection work (§2); injected state is thus always a small fraction of a high-dimensional visual latent. *PhyWorld* (Kang et al. 2025) provides three single-mechanism synthetic domains—*uniform motion*, *parabola*, *collision* (zero/constant/impulsive acceleration)—each with 1,000 ID trajectories (32 frames, 224×224 ; radius/mass $\in [0.7, 1.5]$, velocity $\in [1, 4]$), evaluated on four partitions (ID, r/m-OOD, v-OOD, both-OOD); its single-variable OOD ranges isolate physical extrapolation from data-scale confounds. Two *realistic-simulation* benchmarks (rendered, not camera video) add realism: *Physion* (Bear et al. 2021) for zero-shot object-contact-prediction (OCP) transfer, and *Phy-sion++* (Tung et al. 2023)—ten physical-property scenarios (mass/friction/bounce/deformation)—for direct training with rollout to horizon 64.

Metrics and decision rules. Verdicts use two scale-aware, *external* metrics—latent nMSE ($\|\hat{\mathbf{z}} - \mathbf{z}\|^2$ over target variance) and pixel PSNR of decoded rollouts—external because no injection variant optimizes them directly, and they agree in every case we examined. Latent cosine and linear-probe decodability ρ (Alain and Bengio 2016) are *diagnostics only*: both are duals of the auxiliary losses and rise mechanically when those are optimized, overstating physical competence (we measured cosine $1.50\times$ better while nMSE degraded to $0.21\times$). nMSE fails oppositely—its denominator degenerates when target variance collapses—so parabola is judged on its clean r/m-OOD partition, and all numbers are reported per partition and per horizon. Full pathology catalogue: Appendix F. Splits are trajectory-level; three-seed results report mean $\pm$ sample std.

4.2 Injection Fails in 29 of 30 Cells

Injection degrades or fails to improve prediction in 29 of 30 mechanism-domain cells (17 worse in every seed, 12 within noise; Figure 2). Every cell carries the raw three-seed-mean nMSE alongside the ratio, so the figure doubles as the full result table (per-seed values in Appendix B); read group by group, it answers every design rationale of Table 1. *Hard state injection*: the slot faithfully absorbs position (its supervision loss falls $2.53 \rightarrow 0.015$; slot decodability rises $0.31 \rightarrow 0.96$), yet prediction never improves—worse on parabola and collision ($1.16\times/1.30\times$) and within noise on uniform ($1.19\times$): the constraint acts, and buys nothing. *Soft injection*: the deep-supervision recipe, effective in a low-dimensional state latent where supervised quantities occupy $\sim 38\%$ of the representation (Zahorodnii 2025), does not transfer to a 192-D visual latent where they occupy 2%: it worsens the two both-OOD domains ($1.36\times$ uniform, $1.37\times$ collision) and is within noise on parabola, and combining soft with hard does not rescue ($1.30\times$ on collision). *Correct physics does not help*: the strict $a=g$ head—the ground-truth functional form—worsens uniform

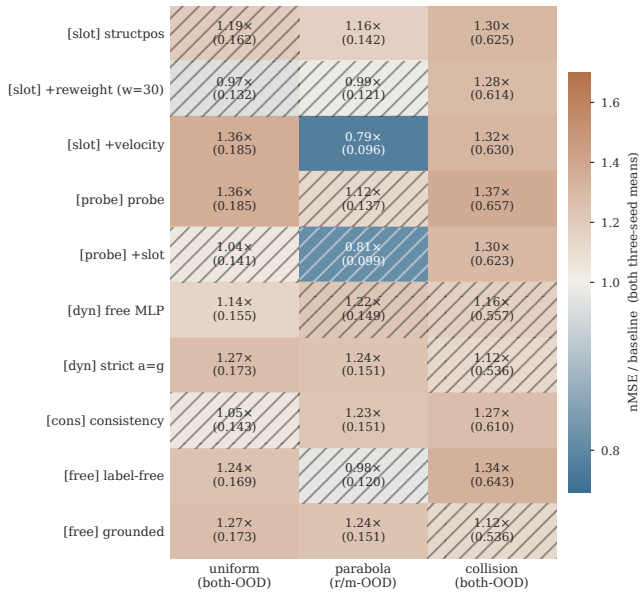


Figure 2: Every injection variant $\times$ domain against the free-rollout baseline (nMSE ratio; every cell and the baseline are three-seed means over seeds 3072/1234/42). Warm = worse, pale = parity, blue = better. *Hatched* cells are within noise—their three seeds overlap the baseline’s, so the ratio alone overstates them; only solid cells separate from the baseline in every seed. Rows follow the three groups of Table 1 in order, and each row label prefixes its mechanism. Injection is at or below baseline in 29 of 30 cells (17 worse, 12 within noise); the collision column is the warmest (1.12–1.37 $\times$, no cell below baseline); the sole gain that separates from baseline in every seed is velocity+slot on parabola (0.79 $\times$, the one solid blue cell). The two dynamics-head rows ($a=g$ and grounded) are the same configuration and print identical values (see text).

by 1.27 $\times$ and parabola to 0.151 nMSE, no better than an unconstrained MLP (0.149). *The purpose-built design does not help its target*: the consistency loss, designed for collision impulses, still degrades collision (1.27 $\times$). *Labels are not the issue*: withholding ground truth changes nothing—the label-free prior and its labeled counterpart both fail to improve any domain in every seed (collision 1.34 $\times$ vs. 1.12 $\times$)—the failure is architectural, not informational.

The sole exception is the mechanism’s signature, not a usable prior: adding *velocity* to the re-weighted slot helps on parabola (r/m-OOD $0.122 \pm 0.007 \rightarrow 0.096 \pm 0.007$, every seed lower) because there velocity is an extrapolable driving variable ($a=g$ makes it linear in time); the same encoding is harmful on uniform (+0.049 over baseline, velocity is a redundant constant) and on collision (+0.017 over the reweighted slot; velocity jumps discontinuously). A prior that requires knowing the domain’s dynamics in advance, and reverses sign across domains, is a diagnostic, not a method—and its best case (−0.026 nMSE) is an order of magnitude below the protocol lever of §4.3.

Domain	Regime	phys. off	phys. on	Δ
uniform	scratch	0.222 ± 0.028	0.657 ± 0.095	+0.435
	fine-tune	0.136	0.173	+0.037
parabola (r/m-OOD)	scratch	0.476 ± 0.169	0.447 ± 0.063	−0.029
	fine-tune	0.122	0.151	+0.029
collision	scratch	0.502 ± 0.043	0.648 ± 0.013	+0.147
	fine-tune	0.479	0.536 ^o	+0.057

Table 2: Injection on/off $\times$ training regime, per domain (latent nMSE $\downarrow$ on the hardest clean split; injected structure: strict $a=g$ head + fixed slot). *Scratch* trains everything from random initialization and is reported over three seeds (mean $\pm$ std); *fine-tune* starts the encoder from pre-trained weights and is read off the three-seed scan (Table 4), so both regimes are three-seed throughout; ^o marks a cell within seed noise there. Δ is the cost of switching injection on within a regime. From scratch, injection is harmful on uniform (Welch 95% CI [+0.276, +0.594]) and collision ([+0.074, +0.220]) and at parity on parabola ([−0.318, +0.260]), whose from-scratch baseline is too seed-unstable (0.343–0.666) to resolve an effect this small. No cell shows injection helping.

From-scratch co-training does not rescue injection.

The remaining defense—structure must be baked in during pretraining—does not hold (Table 2). Co-training from random initialization, over three seeds, leaves injection harmful on uniform (+0.435, Welch 95% CI [+0.276, +0.594]) and collision (+0.147, [+0.074, +0.220]), and at parity on parabola (−0.029, [−0.318, +0.260])—a domain whose from-scratch baseline is itself too seed-unstable (0.343–0.666 across seeds) to resolve an effect of that size. Parity is not a rescue: for this defense to hold, baking the structure in would have to make injection *help*, and in no cell does it. Baking it in from the start also produces the single worst failure in the grid: on uniform the penalty is +0.435 from scratch against +0.037 fine-tuned, an order of magnitude larger. The two regimes trade places on the other two domains, so we do not read a general ordering between them—the robust statement is that neither escapes the harm.

Two corroborations independent of PhyWorld nMSE.

First, *zero-shot transfer* (different dataset, metric, and task): on Physion OCP, no synthetic-trained configuration exceeds the random-encoder prior (0.607 mean AUC over the eight scenarios); the physics-structured variant is the *worst* of all (0.551), below free rollout (0.603), which approaches the prior monotonically over training (0.554 at epoch 1 to 0.603 at epoch 20) without ever crossing it. Per scenario, only Support (+0.10) and Collide (+0.07) clear the random control at all—“more structure, worse transfer,” and most of the apparent zero-shot competence is the architecture, not the learning (Appendix F). Second, *realistic simulation*: trained directly on Physion++, every injection variant we ported (fixed slot, consistency, consistency+acceleration) degrades per-scenario rollout nMSE by 3–10 $\times$ against the identically trained free-rollout model (Appendix B); the gaps dwarf the

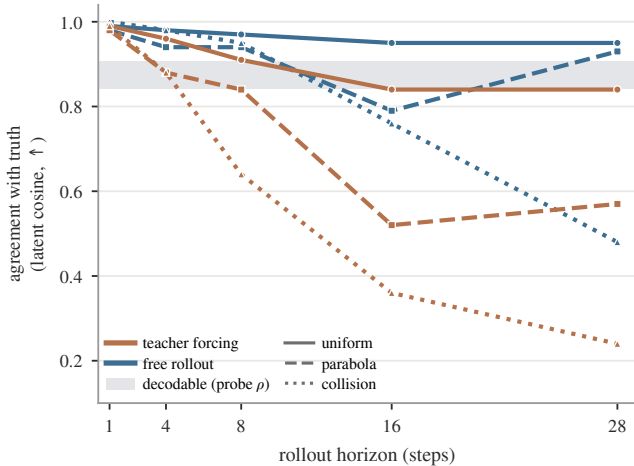


Figure 3: Decodable, used, and still drifting, in all three domains. Colour is the training protocol; line style is the domain. The shaded band is position decodability from the frame embeddings (both-OOD probe ρ , mean over both co-ordinates): the state is *present* to the same degree everywhere (0.84–0.91) and does not decay with horizon. The teacher-forced rollout nevertheless falls out of that band by an amount that tracks dynamics complexity—at horizon 28 it reaches 0.84 on uniform motion, 0.57 on parabola and 0.24 on collision, from 0.98–0.99 at one step. Free rollout (§3.4) recovers much of the gap with no physics added.

three-seed noise band of the baseline. The probe variants have not yet been run on Physion++, so this corroboration covers the slot and consistency mechanisms only.

4.3 Mechanism: Load-Bearing but Redundant

The puzzle of §1 localizes here: in all three domains, position is linearly decodable from the latent at every horizon, yet the rollout drifts away from it, and the further the harder the dynamics (Figure 3). Decodability alone would not settle what the model does with that state—probe-recoverable information need not be information the model uses (Elazar et al. 2021)—so we measure four things: that the redundant route *exists* (Test 1), that the constraint *acts* and competes with prediction (Test 2), that its strength is *not* what limits it (Test 3), and, by intervening on trained models, which channel prediction actually reads (Test 4). Test 4 refutes the reading we first defended: the injected slot is not bypassed at all.

Test 1: the physical constraint is injected. Probing *only the 190 non-slot dimensions* decodes position as well as the full latent in all three domains, undiminished by pinning the slot, while a random-2-dimension control fails (Table 3, top). Position is thus a redundant, distributed copy: a route exists that never passes through a structured slot. This establishes that the route is *available*; that prediction takes it is the reading we defend below, not a measurement—linear decodability need not imply use (Elazar et al. 2021), and the probe here reads frame embeddings, not rolled-out latents.

	uniform	parabola	collision
<i>The bypass: position decodability ρ (both-OOD)</i>			
full 192-d latent	0.92	0.85	0.78
black box only (190 d)	0.92	0.85	0.79
black box, slot pinned	0.92	0.86	0.79
the pinned slot itself (2 d)	0.92	0.85	0.51
random-2-d control		0.2–0.5	
<i>The cost: what injection does to the encoder</i>			
loss ratio ($\lambda \mathcal{L}_{\text{phys}} / \mathcal{L}_{\text{pred}}$)	44×	125×	15×
effective dim, $\lambda=1 \rightarrow 10$	41 $\rightarrow$ 4	13 $\rightarrow$ 3	28 $\rightarrow$ 17
collapse	–90%	–75%	–39%

Table 3: Why injection cannot help: the black box already carries position, and forcing it elsewhere costs the encoder. *Top*—the 190 non-slot dimensions decode position as well as all 192, and pinning position into a slot does not diminish that copy, so an injected slot is a second carrier of a state the predictor already had; a random-2-d control fails, so the redundancy is distributed rather than sparse. On collision the pinned slot is the *only* degraded read (0.51), yet the black box is untouched (0.79)—the duplication at its clearest. *Bottom*—the constraint nevertheless dominates optimization (weighted physical loss 15–125× the prediction loss at $\lambda=10$, a gradient-magnitude proxy) and collapses the encoder’s participation ratio by 39–90%. Acting strongly on a quantity the latent already encodes is exactly the failure mode.

Test 2: the constraint acts harmfully. Both measurements move, which indicates the constraint acts (Table 3, bottom): at probe weight $\lambda=10$ the weighted physical loss outweighs the prediction loss by one to two orders of magnitude (a gradient-magnitude proxy), and the encoder’s effective dimensionality collapses with it. At $\lambda=1$, the setting of §4.2, the ratio is milder (3–20×) yet the harm persists. The constraint demonstrably reshapes the representation—in a direction that competes with prediction.

Test 3: re-weighting the physical loss does not help. Sweeping the slot weight $w=1 \rightarrow 300$ across every partition (Appendix B) moves the harm systematically—the mechanism’s direction is confirmed—but weighting rescues only 2 of 4 domain–partition decision cells, to *parity* and never a net gain: uniform both-OOD returns exactly to baseline (0.132 ± 0.017 vs. 0.136 ± 0.008), uniform r/m-OOD never recovers, and collision worsens monotonically with weight. Insufficient physical loss is therefore not the root cause. What the sweep cannot fix is what stays constant across it—the redundant route through the black box—which is why we read the limit as architectural, and why the extrinsic port of §3.5, which removes that route by construction, is the informative test.

Test 4: the injected slot is load-bearing—and that is the point. Intervening on trained models overturns the natural reading of Tests 1–3. *Steering*: offsetting the position the slot decodes moves the model’s own black-box state by 1.5–2.0× the offset, while a norm-matched random direction moves it by ≈ 0 (–0.03 to +0.01); the decomposition

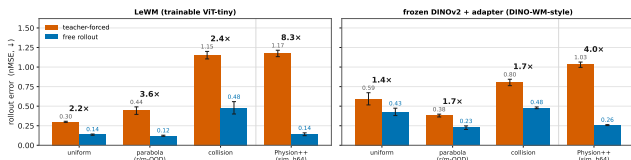


Figure 4: Teacher forcing vs. free rollout, both backbones on one axis (hardest clean split, nMSE ↓; solid = LeWM, hatched = the frozen-DINOv2 variant of §3.5; fold-change above each pair). Bars are three-seed means, error bars one sample std over seeds 3072/1234/42: every teacher-forcing/free-rollout interval is disjoint. The physics-free protocol switch is worth 2.2–8.3× on LeWM and 1.4–4.0× on frozen DINOv2—the lever is backbone-independent, and largest where the dynamics are hardest (collision, Physion++).

is additive to within 3%, so the perturbation stays in the linear regime. *Counterfactual patching*: replacing the slot with a donor trajectory’s makes prediction follow the donor 47–72% of the way, where the same operation on a model with no injected slot—two dimensions of no physical meaning—moves it 1.5–11%. *Jacobians*: differentiating the predicted position with respect to each latent coordinate, no intervention designed, ranks the two injected dimensions 1st and 2nd of 192 by sensitivity, against ranks ~99 for the same two dimensions before injection. Three methods with disjoint failure modes agree, over three seeds and both backbones (Appendix D).

So prediction does not route around the slot; it reads it. What the same interventions also show is that the black box keeps its own copy: pinning the slot leaves black-box decodability undiminished (Table 3), and the black box still holds 96–99% of total Jacobian sensitivity by sheer dimension count. Both channels are load-bearing, and prediction error does not move—which is what the redundancy hypothesis predicts and what a starved slot would not. The injected slot is *load-bearing but redundant*: it carries load, and the load it carries is a duplicate.

Replacing teacher forcing with free rollout cuts the hardest-split error 2.2–3.6× across the three domains and 8.3× on Physion++, every teacher-forcing/free-rollout interval disjoint over three seeds, and the whole pattern replicates on the frozen-DINOv2 backbone (Figure 4; per-domain values in Appendix C). Crucially the gain is not an OOD patch: *every* partition improves, including ID (2.0–4.6×). The horizon must match dynamics complexity: collision improves markedly at $H \approx 20$ ($0.393 \rightarrow 0.294$) while smooth domains degrade with excess horizon (uniform horizon-28 cosine $0.969 \rightarrow 0.814$ at $H=16$); realistic-simulation scenes have the largest appetite—extending $H=8 \rightarrow 28$ (with geometric scale jitter stabilizing amplitude; Appendix E) drives horizon-64 nMSE from 0.280 to 0.014, a ~19× reduction with no inflection found (Appendix E).

As the positive control of §3.4, this result carries the methodological weight of the paper: on the same code, metrics, seeds, and data that judged all thirty injection cells, a

physics-free intervention produces multiples—so the injection failure of §4.2 is attributable to the injected structure, not to the pipeline or the yardstick.

4.4 External Control: Extrinsic Is Necessary, Not Sufficient

Extrinsic port. The faithful extrinsic port (PIWM; §3.5) learns the correct physics and, where its encoder is in-distribution, beats our shared-latent model: rolled-out position ρ reaches 0.96 (ID) and 0.97 (v-OOD) vs. LeWM’s 0.93/0.87. But under radius/mass shift—which changes appearance, hitting the VAE encoder—it collapses to $\rho = 0.33$ (r/m-OOD) and 0.48 (both-OOD) while LeWM holds 0.89/0.87; appearance-changing shift is an axis its original case studies do not vary. The architectural prediction of §4.3 is confirmed from the outside: making the physical state load-bearing by construction is what extrinsic designs buy (their equations alone do not transfer, §4.2); yet closing the bypass leaves the encoder-OOD problem unsolved. Extrinsic structure is necessary but not sufficient.

Cross-backbone replication. On the frozen-DINOv2 variant every headline finding replicates across three seeds. Presence is even stronger—the physics-naïve frozen encoder decodes position at $\rho=0.95$ (vs. 0.90 for LeWM), with the same redundant black-box copy—so both the presence of the state and its duplication across the latent are properties of generic visual features, not of our training. The protocol lever transfers (1.4–1.7× synthetic, 4.0× Physion++; hatched bars in Figure 4). Injection still never wins (27/30 at or below baseline; Appendix C); the three above-parity cells dissolve under content-free controls—a shuffled target, or a 10×-weaker pin at matched loss weighting, keeps the apparent gain while ID error degrades, i.e. capacity-restriction regularization, not physics (Trap 5, Appendix F). One difference: with the encoder frozen, injection is inert rather than harmful—the losses reshape only the adapter, bounding damage but foreclosing benefit. Across every control, the same sentence survives: physical state in a latent world model is *load-bearing but redundant*.

5 Conclusion

We asked whether injecting physics can make a latent world model physically consistent, and answered with a controlled anatomy: injection fails in 29 of 30 mechanism–domain cells—with the exact gravitational form, with perfect labels, and whether grafted onto a pretrained encoder or co-trained from scratch. The account we defend is that the state is already there: the non-slot dimensions carry the same position ($\rho = 0.79$ –0.92), so an injected copy adds no signal while the constraint consumes capacity and gradient—*load-bearing but redundant*. Intervention settles the half we could previously only infer, and settles it against our first reading: the injected slot *is* read, by three methods that fail in different ways, while the black box keeps its own copy of the same state. Both of the account’s predictions held: re-weighting the slot reaches only parity, and an architecture that removes the redundant route does help, until an OOD shift reaches its encoder. Removing that route is necessary, not sufficient.

What moved prediction was never the structure but the training protocol, worth $2.2\text{--}8.3\times$ on identical code and metrics.

The constructive reading: an inductive bias must supply what the latent is missing—*how the state evolves, not what the state is*. Two routes stay open: injections that target dynamics or conserved quantities rather than state, and extrinsic designs paired with OOD-robust encoders. Our own evolution-targeting variants are deliberately simple: a second-order kinematic update and a finite-difference consistency prior, written onto slot coordinates. Their failure shows that pinning the evolution this way is not *automatically* useful—not that evolution injection fails in general. Collision makes the gap concrete: it turns on contact, impulse and inter-object relation, none of which a smoothness prior over one object’s positions can express, so our collision result bounds that prior rather than the operator-level designs (symplectic, Hamiltonian, learned contact) that would represent it properly.

Limitations. Our training-dynamics conclusions rest primarily on one compact backbone ($\sim 10\text{M}$ ViT-tiny JEPa); the headline findings replicate on a frozen-DINOv2 backbone (§4.4), where injection is inert rather than harmful—so the *harm* claim is established on the trainable-encoder regime. All 30 scan cells, the re-weighting sweep, and the from-scratch 2×2 are three-seed; the Physion++ variants remain single-seed. The probe variants remain untested on Physion++, and the one favorable cell (velocity on parabola) suggests dynamics-matched injection deserves study we did not perform. On the mechanism, the half we can now demonstrate is that the injected slot is read; the half that stays partly inferential is the black box’s own causal role. Amnesic projection, the standard tool for that question, cannot answer it here: erasing position linearly requires removing 116–154 of the 190 non-slot dimensions, and at that rank removing position is indistinguishable from removing the same number of dimensions at random—a direct consequence of how distributed the copy is, and a limit we quantify rather than a step we skipped. Wholesale replacement of the black box does establish its causal role, but its controls pass in only one of six backbone–domain cells. So “both channels carry the same state” rests on measurement for the slot and on convergent but incomplete evidence for the black box.

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A Training and Evaluation Details

Reproducibility. All training configurations, per-run logs, evaluation protocols, and the exact numbers behind every table and figure are archived alongside the code; the study builds on public assets (LeWM, PhyWorld, Physion, Physion++). We will release the evaluation suite, including the per-horizon/per-partition audit tools and the random-encoder controls.

Hyperparameters. All PhyWorld runs: AdamW, lr 5×10^{-5} (from-scratch reruns: 2×10^{-5}), weight decay 10^{-3} , bf16, gradient clip 1.0, 20 epochs; the from-scratch cells of the pretraining-regime 2×2 train longer and at a lower learning rate so their baselines converge cleanly—60 epochs on uniform motion and collision, 120 on parabola, which needed the extra budget—with injection on and off sharing the identical budget within each domain, so every Δ in Table 2 is a within-domain comparison. Batch size 64 for free rollout ($H=8$), 128 for teacher forcing ($H=1$), 48 for $H \geq 16$. History size 3; embedding dimension 192; SIGReg weight 0.09. Seeds: 3072/1234/42 where seed-replicated; both arms of the pretraining-regime 2×2 are three-seed (Table 2): the from-scratch cells were replicated directly, and the fine-tune cells are read off the three-seed scan of Table 4, which is the same regime. Physion++ runs use the readout split (800 clips) with the same optimizer.

OOD partitions. ID: radius/mass $\in [0.7, 1.5]$, velocity $\in [1.0, 4.0]$. OOD draws radius/mass from $[0.3, 0.6] \cup [1.5, 2.0]$ and velocity from $[0, 0.8] \cup [4.5, 6.0]$. Collision uses the four-column variant (two objects’ masses and velocities). Training sets are ID-only (1,000 trajectories); evaluation covers all four partitions ($\geq 1,600$ trajectories per domain).

Probe protocol. Ridge regression ($\alpha=1$), trajectory-level 80/20 splits, probing the encoder output (not the projector), pretrained “paper” initialization, and $K=4$ stacked consecutive latents for velocity. Each choice matters: the naive combination (random init, single frame, through-projector) reads $\rho=0.166$ for uniform velocity where the corrected protocol reads 0.939 (Trap 4, Appendix F). Random-encoder and pixel-statistics controls are run alongside every probe.

From-scratch 2×2 , per seed. Hardest clean split, latent nMSE (uniform/collision both-OOD, parabola r/m-OOD), seeds 3072/1234/42: uniform off 0.192/0.230/0.246, on 0.750/0.560/0.662; parabola off 0.343/0.420/0.666, on 0.375/0.494/0.473; collision off 0.538/0.513/0.454, on 0.635/0.661/0.649. The parabola from-scratch baseline spans 0.343–0.666 across seeds—a spread wider than the effect being measured, which is why that cell is reported as parity rather than as a gain.

B Full Injection Results

Table 4 is the numeric form of Figure 2, with the seed annotations spelled out.

Probe/slot factorial (uniform, free rollout). Latent both-OOD nMSE / pixel both-OOD PSNR (dB): baseline 0.131 / 20.41; probe-only 0.167 / 19.83; slot+reweight 0.114 /

	uniform (both-OOD)	parabola (r/m-OOD)	collision (both-OOD)
Free-rollout baseline	0.136	0.122	0.479
Fixed slot	0.162°	0.142	0.625
+reweight ($w=30$)	0.132°	0.121°	0.614
+velocity	0.185	0.096	0.630
Probe	0.185	0.137°	0.657
+slot+reweight	0.141°	0.099°	0.623
Kinematic (free MLP)	0.155	0.149°	0.557°
Kinematic (strict $a=g$)	0.173	0.151	0.536°
Consistency	0.143°	0.151	0.610
Label-free prior	0.169	0.120°	0.643
Grounded prior	0.173	0.151	0.536°

Table 4: The 30-cell scan (hardest clean split, latent nMSE $\downarrow$; every cell and the baseline are three-seed means over seeds 3072/1234/42). **Bold** marks the one seed-robust gain (velocity+slot on parabola: 0.096 ± 0.007 vs. baseline 0.122 ± 0.007 , intervals disjoint). ° marks cells within noise—their three seeds overlap the baseline’s, so the cell does not separate from baseline in either direction; unmarked cells are worse in every seed. 17 cells worse, 12 within noise, 1 better. Kinematic (strict $a=g$) and the grounded prior are the same configuration—a fixed slot with a learned-constant kinematic head under ground-truth labels—so their rows are identical. That row is the one exception to the three-seed rule above: because the configuration had been launched twice under both names before the duplication was noticed, it has two independent seed-3072 runs, and we average all four (both 3072 draws plus the 1234 and 42 top-ups) rather than discard one. The two 3072 draws bracket the mean (e.g. uniform 0.206 and 0.166), so the extra weight on that seed does not bias the cell in either direction.

21.30; probe+slot+reweight 0.125 / 20.02. The combination underperforms the slot alone on both scales; the latent and pixel verdicts agree.

Slot reweighting sweep (uniform). Slot weight 1/30/100: 0.183 / 0.114 / 0.136 (single seed), with the $w=30$ cell seed-replicated at 0.132 ± 0.017 against the baseline 0.136 ± 0.008 . Kinematic head on the reweighted slot: velocity-OOD decoded position $\rho=0.991/0.987$ (x/y); uniform pixel horizon-28 23.34 vs. 22.09 dB (+1.25); appearance-OOD remains the weak axis (decoded ρ $0.29 \rightarrow 0.43$, vs. 0.96 without the kinematic head).

Bypass probe details. Protocol for §4.3, which reports the numbers. Two models are probed: the free-rollout baseline (no injection) and the slot-pinned model (structpos, $w=30$). For each, a ridge probe is fit on the dimension subset alone—the full 192, the black box [2:192], the slot [0:2], or a random pair as control—with train/test trajectories split 80/20, so decodability is measured out of sample. Values are Pearson ρ against ground-truth position on the both-OOD partition, averaged over both coordinates, from frame embeddings rather than rolled-out latents. The slot absorbs position (its supervision loss falls $2.53 \rightarrow 0.015$) without displacing the black box’s redundant copy.

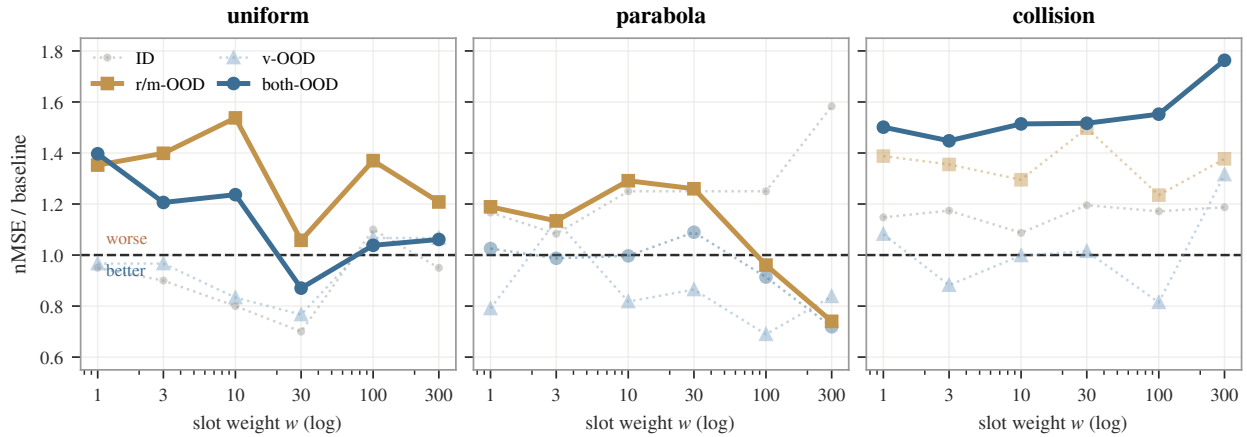


Figure 5: Dose-response re-weighting, all partitions. Solid curves are the four decision cells (hardest clean split); dotted are partitions we do not judge on—ID is not an extrapolation test, and parabola’s v-OOB/both-OOB nMSE is denominator-degenerate (Appendix F). Only 2 of the 4 decision cells reach parity (uniform both-OOB, parabola r/m-OOB); uniform r/m-OOB and collision both-OOB stay above baseline at every weight, and collision degrades as weight grows. The sweep is single-seed; both parity cells were re-run over three seeds and land at baseline (§4.3).

Encoded quantity vs. domain (position monotone, velocity sign-flipping). The reweighted slot’s effect is systematic in what it encodes (hardest clean split; baseline: uniform both-OOB 0.131, parabola r/m-OOB 0.127, collision both-OOB 0.393):

slot content	uniform	parabola	collision
position (structpos)	0.132	0.160	0.596
position+velocity	0.207	0.096	0.621
Δ (adding velocity)	+0.075	−0.064	+0.025

Position is an output variable with the same role in every domain, so pinning it degrades monotonically with dynamics complexity and never flips sign. The marginal effect of adding velocity (+0.075/−0.064/+0.025) tracks velocity’s role in each domain—redundant constant (uniform), linear driver (parabola), discontinuous jump (collision). The parabola gain ($0.122 \pm 0.007 \rightarrow 0.096 \pm 0.007$, every seed lower) is thus the sole seed-robust structural gain in the grid: a fixed form landing on a matching domain, not a portable prior.

Physion++ per-scenario (all ten scenarios, FR at 20 epochs). Rollout nMSE: bouncy_platform 0.041, bouncy_wall 0.094, friction_cloth 0.071, friction_collision 0.062, friction_platform 0.041, mass_collision 0.083, mass_dominoes 0.058, mass_waterpush 0.122; deformable scenes are the common hard case for every method (deform_clothhang 1.375, deform_clothhit 0.772). Injection variants degrade the rigid-body scenarios by 3–10 $\times$ (Table 5) while leaving the deformable failure in place.

Physion++ horizons. Overall nMSE at horizons 16/32/64: baseline ($H=8$) 0.016/0.140/0.280; $H=20$ 0.022/0.033/0.220; $H=16$ +scale 0.010/0.035/0.136; $H=20$ +scale 0.012/0.024/0.087; $H=28$ +scale reaches horizon-64 nMSE 0.014. Free rollout is the main long-horizon lever (horizon-32 cosine 0.91 vs. 0.79–0.87 for

Scenario	FR	+slot	+cons.	+cons.+acc.
friction_platform	0.041	0.119	0.078	0.067
mass_collision	0.083	0.510	0.682	0.696
mass_dominoes	0.058	0.452	0.586	0.597
mass_waterpush	0.122	0.363	0.663	1.476

Table 5: Physion++ direct training: per-scenario rollout nMSE ($\downarrow$) at 20 epochs. Injection degrades realistic-simulation prediction by 3–10 $\times$ on the mass/friction scenarios shown (deformable-object scenes are the hardest category for all methods, including the baseline). Single-seed; the gaps far exceed baseline seed noise.

every physics-augmented variant); geometric scale jitter acts as an amplitude stabilizer for long rollouts: without it, $H=20$ explodes on friction-collision (nMSE 24.6 at cosine 0.99); with it, 0.019.

C Cross-Backbone Replication: Frozen DINOv2

The cross-backbone variant (§3.5) freezes a DINOv2-small encoder (Oquab et al. 2024)—in the style of DINO-WM (Zhou et al. 2025)—and trains only the projector (as adapter) and the identical predictor stack, losses, and seeds (~ 130 runs). Free-rollout baselines (three-seed): uniform both-OOB 0.427 ± 0.039 , parabola r/m-OOB 0.225 ± 0.019 , collision both-OOB 0.479 ± 0.009 ; Physion++ horizon-64 0.258 ± 0.005 vs. teacher forcing 1.030 ± 0.028 .

Teacher forcing vs. free rollout, per domain. Values behind Figure 4 (hardest clean split, latent nMSE $\downarrow$, three-seed mean $\pm$ sample std). LeWM: uniform $0.300 \pm 0.007 \rightarrow 0.136 \pm 0.008$ (2.2 $\times$), parabola $0.443 \pm 0.048 \rightarrow 0.122 \pm 0.007$ (3.6 $\times$), collision $1.152 \pm 0.048 \rightarrow 0.479 \pm 0.079$ (2.4 $\times$), Physion++

horizon-64 $1.174 \pm 0.041 \rightarrow 0.141 \pm 0.018$ ($8.3\times$). Every teacher-forcing interval is disjoint from its free-rollout counterpart. The frozen-DINOv2 counterparts above give $1.4\text{--}4.0\times$ over the same four settings, so the lever survives replacing a trainable ViT-tiny with a frozen general-purpose encoder.

Injection scan on the frozen-DINOv2 backbone
(nMSE / free-rollout baseline; vermilion = worse)

[slot] structpos	$1.17 \times \dagger$ (0.502)	$1.02 \times \dagger$ (0.229)	$0.97 \times \dagger$ (0.463)
[slot] +reweight	$0.81 \times \dagger$ (0.347)	$1.05 \times \dagger$ (0.236)	$1.01 \times \dagger$ (0.485)
[slot] +velocity	$0.93 \times \dagger$ (0.398)	$1.01 \times \dagger$ (0.227)	$0.99 \times \dagger$ (0.473)
[probe] probe	$1.02 \times \dagger$ (0.434)	$1.03 \times \dagger$ (0.231)	$1.03 \times \dagger$ (0.493)
[probe] +slot	$0.91 \times \dagger$ (0.390)	$1.10 \times \dagger$ (0.247)	$1.02 \times \dagger$ (0.487)
[dyn] free MLP	$0.99 \times \dagger$ (0.423)	$1.04 \times \dagger$ (0.234)	$1.11 \times \dagger$ (0.530)
[dyn] strict $a=g$	$1.08 \times \dagger$ (0.459)	$1.16 \times \dagger$ (0.261)	$1.27 \times \dagger$ (0.610)
[cons] consistency	$1.00 \times \dagger$ (0.428)	$1.01 \times \dagger$ (0.227)	$0.96 \times \dagger$ (0.461)
[free] label-free	$0.96 \times \dagger$ (0.411)	$1.00 \times \dagger$ (0.224)	$1.07 \times \dagger$ (0.512)
[free] grounded	$1.11 \times \dagger$ (0.473)	$1.13 \times \dagger$ (0.254)	$1.27 \times \dagger$ (0.607)
	uniform (both-OOD)	parabola (r/m-OOD)	collision (both-OOD)

Figure 6: The injection scan repeated on the frozen-DINOv2 backbone (nMSE over the free-rollout baseline; mean over available seeds, $\dagger$ = three seeds). 10 cells worse / 17 parity / 3 better (27/30 at or below baseline). The dynamics-head variants are the worst here too (strict $a=g$ and the grounded prior reach $1.27\times$ on collision); the three above-parity cells, all on uniform, are the re-weighting confound of Trap 5.

Presence and bypass. The frozen encoder, which has never seen PhyWorld and received no physics supervision or gradient, decodes position from real-frame embeddings at $\rho=0.951$ (both-OOD)—*higher* than the PhyWorld-trained LeWM encoder (0.899). The 190 non-slot dimensions alone decode at 0.951, indistinguishable from the full latent; pinning the slot at weight 300 leaves the black-box copy at 0.973. Presence and the bypass are properties of generic visual representations.

Inert, not harmful. Unlike the trainable-encoder regime, the core variants (re-weighted slot, probe, consistency) land within the baseline’s seed band on all three domains, while the plain slot ($1.42\times$ on uniform) and the dynamics-head variants ($1.15\text{--}1.29\times$ on all domains) still hurt. With the encoder frozen, injection losses can only reshape the $384 \rightarrow 192$ adapter: the bypass cannot be displaced, which bounds the damage—and forecloses the benefit. Injection-variant variance also roughly doubles (± 0.087 vs. baseline ± 0.039 on

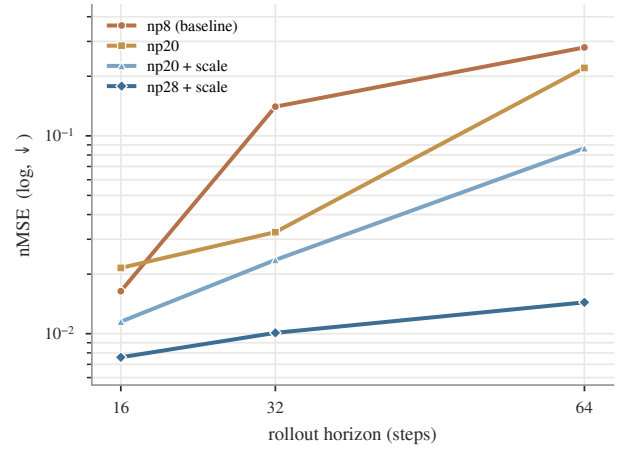


Figure 7: Physion++ direct training: per-horizon rollout nMSE (log scale) as the training horizon grows ($H=8 \rightarrow 28$, +scale jitter). Long-horizon error falls monotonically to $1/19$ of baseline with no inflection: realistic-simulation dynamics reward long-rollout training more than the synthetic domains do.

uniform), so single-seed cells on this backbone are uninformative.

D Causal Interventions on Trained Models

Protocol and full results for Test 4 (§4.3). All three interventions run at inference on existing checkpoints—no retraining—over three seeds (3072/1234/42), three domains, and both backbones. Position is read with the same ridge probe used elsewhere, fit on the training split.

Why the readout is the black box. Each intervention perturbs one channel and reads position off the *other*. Reading back the channel just written is partly trivial: the predictor copies its input forward, so the slot readout moves whether or not the perturbation reached the computation. The black-box readout is the one that reflects it (it is also what drives the pixel decoder). Predictions are recorded *before* the per-step perturbation is applied, so every number below is read off what the model itself produced.

Steering. For a target offset δ (in units of position std), we add the minimum-norm latent offset that moves the named channel’s decoded position by δ —for the supervised slot, writing $\text{pos} + \delta$ directly, which is better conditioned—and report $g = \Delta(\text{decoded position})/\delta$. Slot gains on the black-box readout, three-seed mean $\pm$ std, against a norm-matched random direction in the black box:

backbone	domain	g_{slot}	random control
LeWM	uniform	$+1.52 \pm 0.03$	-0.017
LeWM	collision	$+2.01 \pm 0.20$	-0.003
LeWM	parabola	$+1.15 \pm 1.40$	-0.017
DINOv2	uniform	$+1.69 \pm 0.06$	-0.011
DINOv2	parabola	$+0.59 \pm 0.30$	$+0.001$
DINOv2	collision	$+1.94 \pm 0.31$	-0.031

Additivity holds to within 3% ($g_{\text{slot}} + g_{\text{bb}}$ vs. steering both together), so the perturbation stays in the linear regime. *The magnitudes are not proportions*: values above 1 reflect probe $R^2 < 1$, an unperturbed action input, and the 3-frame history anchor. They support “used vs. not” and “this channel vs. that one”, nothing quantitative. LeWM/parabola is excluded: its seeds read $-0.10/+0.90/+2.66$, a spread wider than the effect, and a $20\times$ sweep of δ leaves the per-seed values essentially unchanged, so this is three different models rather than one noisy measurement.

Counterfactual patching. We replace the slot with a donor trajectory’s slot at the same timestep, leaving the black box at its own values, and regress the decoded position on [own truth, donor truth]; the follow-fraction is the donor’s share. The control is the same operation on a model with no injected slot, where those two dimensions carry no physical meaning: LeWM/uniform $0.015 \rightarrow 0.468$, DINOv2/uniform $0.112 \rightarrow 0.650$, DINOv2/collision $0.091 \rightarrow 0.721$. The LeWM parabola and collision controls are not clean (a no-slot model follows $0.42\text{--}0.51$), and we exclude those cells; a collinearity explanation fits parabola (pooled own–donor correlation 0.76) but not collision (0.22), so the cause is unresolved.

Per-dimension Jacobians. $J[c, j] = \partial(\text{decoded position}_c)/\partial(\text{latent}_j)$, averaged over 400 frames, reported per dimension so a 2-d slot and a 190-d black box share a scale. Gradient sensitivity is standard (Simonyan, Vedaldi, and Zisserman 2013; Sundararajan, Taly, and Yan 2017), here applied to an intermediate representation rather than to pixels. Unlike the interventions, it chooses no direction, which matters because a redundant code barely moves along any single one; all six cells are usable here where the black-box replacement below leaves one. Its known weakness is the reason we do not rest on it alone: a gradient is local and linear, so it reports what a small change would do rather than what changing the channel does, and some saliency methods are insensitive to the model they purport to explain (Adebayo et al. 2018). The baseline rows answer that second concern directly—the readout tracks whether the model was injected, not the input structure—and the interventions above answer the first. The baseline rows are a built-in control: with no injection those two dimensions are arbitrary and should rank mid-pack, and they do (ratio $0.80\text{--}0.98$, ranks $\sim 90\text{--}162$ of 192). After injection they rank 1st–2nd on LeWM uniform/parabola and DINOv2 uniform, at $2.3\text{--}4.0\times$ the per-dimension sensitivity of the black box; DINOv2/parabola is the exception (ratio 0.99 , ranks $111/90$). The slot nonetheless holds only $0.8\text{--}4.0\%$ of *total* sensitivity—two dimensions against 190—which is the analytic form of “both channels carry load”.

Amnesic projection: a limit we quantify. We remove the linearly position-predictive subspace from the black box by iterative nullspace projection (Ravfogel et al. 2020), with a rank-matched random-ablation control. Erasing position requires removing 116–154 of the 190 dimensions—the hardest quantification of redundancy in this paper, and far

stronger than the random-2-dimension control of Table 3. But at that rank the test loses selectivity: in five of six cells, removing position damages the rollout *less* than removing the same number of dimensions at random (e.g. uniform structpos 0.626 vs. 1.099). The test cannot discriminate in this representation, which follows from how distributed the copy is rather than from how we ran it. Two caveats we state rather than smooth over: the control is matched in rank but not in variance (INLP selects linearly predictive directions, which may be lower-variance than random ones), and an earlier run stopped at 24 removed dimensions showed position-removal damaging the rollout *less* than random removal—apparent support for a bypass—a contrast that disappears once INLP is run to convergence.

Black-box replacement. Replacing k black-box dimensions outright with a donor’s (slot untouched) sidesteps the min-norm problem: on a baseline model, where the black box is the only route position can take, the min-norm patch yields a follow-fraction of 0.089 while replacing all 190 yields 0.961 —an order of magnitude apart on the same model and channel. Sweeping k gives the crossing point at which moving the black box buys as much as moving the 2-d slot. A cell is usable only if its baseline both ignores a patch to the two dimensions it does not have and follows the donor once the whole black box is replaced; only LeWM/uniform passes (0.015 and 0.961), where the crossing sits near 100 dimensions. The remaining five cells fail one or both controls and their crossings are not quoted.

E Augmentation: a Domain-Specific Lever with a Reversal Boundary

Augmentation is not part of the paper’s main line (it is neither an injection mechanism nor a protocol variable), but its boundary behavior is a useful caution. On synthetic domains, per-trial appearance jitter (brightness/contrast, strength 0.5) roughly halves the hardest split: uniform $0.131 \rightarrow 0.068$ (-48%), parabola -63% (r/m-OOD $0.127 \rightarrow 0.065$); geometric scale jitter, only in combination with a long horizon, sets the collision record ($0.393 \rightarrow 0.172 \pm 0.004$ over three seeds). Interactions are non-monotone: appearance conflicts with long-horizon training (0.472 , worse than either alone), scale synergizes (0.208), the triple is worse than the best pair (0.253); temporal-stride augmentation loses even to appearance on velocity-OOD (0.556 vs. 0.376), leaving unseen velocities the open hard case.

The appearance recipe reverses on realistic simulation: the same jitter that halves synthetic OOD error degrades Physion++ friction-collision prediction by two orders of magnitude (nMSE $0.062 \rightarrow 6.44$) while the *aggregate* long-horizon cosine improves (Trap 1 below), and it does not help zero-shot transfer either (0.597 , below the 0.607 random-prior ceiling). The boundary is principled: appearance jitter encodes an appearance-invariance assumption that holds under PhyWorld’s simple renderer but breaks where appearance carries physical information (friction, mass, material). Scale jitter, which assumes only size-invariance, keeps helping at realistic-simulation scale (Appendix B); appearance

jitter flips sign. Augmentation gains are domain- and type-specific, not portable.

F Metric Pathologies and Evaluation Traps

Our anatomy repeatedly reached conclusions opposite to those suggested by common metrics; the decision rules of §4.1 distill the following five failure modes, each of which reversed at least one finding.

Trap 1: cosine and probe metrics are duals of the training loss. Probe correlation and latent cosine rise mechanically when probe or slot losses are optimized—they measure whether information is *present*, not whether prediction *uses* it. Probe supervision lifts velocity decodability from $\rho=0.44$ to 0.80 and gives the most stable long-horizon *cosine* (0.882 vs. 0.843), while its *pixel* rollout is the worst in the comparison (19.93 vs. 20.64 dB at horizon 28). Appearance augmentation on Physion++ improves the aggregate long-horizon cosine while per-scene nMSE collapses up to $100\times$ (deform_clothhit: cosine $0.610 \rightarrow 0.913$, nMSE $0.772 \rightarrow 3.69$). Judging either result by direction-only or readout metrics inverts the conclusion. Our own early sweep was misled this way: judged on $K=4$ probe ρ , $\lambda=50$ “won”; re-judged on prediction loss, the ranking reversed.

Trap 2: nMSE denominators can degenerate. When the target sequence degenerates (a parabola ball exits the frame, target variance $\rightarrow 0$), per-horizon nMSE explodes— 3×10^4 – 2×10^6 at horizon ~ 28 across all six parabola baseline variants, while cosine sits at a healthy 0.55–0.95—and pollutes every aggregate that includes late horizons. This is why parabola is judged on its clean r/m -OOD partition throughout. Traps 1 and 2 point in opposite directions (cosine under-reports failure; degenerate nMSE over-reports it); neither metric family is safe alone, hence per-partition, per-horizon cross-validation.

Trap 3: zero-shot transfer is bounded by the architecture prior. On Physion OCP, a *randomly initialized* encoder scores mean AUC 0.607. No synthetic-training configuration exceeds it: free rollout approaches it (0.603), physics variants fall below it (0.551–0.582), appearance augmentation 0.579–0.597; training longer approaches the ceiling monotonically (0.554 at epoch 1 to 0.603 at epoch 20) without crossing. Per-scenario, only Support (+0.10) and Collide (+0.07) show signal above the random control. Zero-shot claims must be calibrated against the random-prior floor, or they measure the architecture, not the learning.

Trap 4: protocol confounds fabricate results in both directions. Three probe-protocol choices (random vs. pre-trained initialization, single-frame vs. stacked probes, probing through the projector) multiply into a false negative: the naive stack reads uniform velocity decodability at $R^2=0.166$, the corrected protocol 0.939. A silent initialization bug (only 24 of 216 loadable encoder keys actually loaded) invalidated an entire 45-configuration sweep before a loading guard caught it. And converged training loss proves nothing about rollout: our clean from-scratch baseline converges to the pretrained model’s prediction loss (0.006 vs. 0.005) while its rollout OOD error is $2.8\times$ worse.

Trap 5: loss re-weighting masquerades as physics. On the frozen-DINOv2 backbone, heavily re-weighting the pinned position slot (weight 300; the 2 slot dimensions then carry 76% of the prediction loss) improves uniform both-OOD from 0.427 ± 0.039 to 0.286 ± 0.015 —apparently the one clear injection win on either backbone. Two content-free controls dissolve it: pinning the slot to a *shuffled*, physically meaningless target at the same weight reaches 0.330 ± 0.013 , and weakening the pin $10\times$ while keeping the weighting reaches 0.329 ± 0.084 —most of the gain, with no usable physics. All three variants degrade in-distribution error ($0.099 \rightarrow 0.115$ – 0.136), the signature of capacity restriction (the predictor is effectively reduced to a 2-dimensional task), and the same recipe *worsens* collision ($1.11\times$) and parabola ($1.14\times$). An apparent injection gain must be checked against a content-free control at matched loss weighting.

Auxiliary-loss dominance (measurement behind §4.3, Test 2). At probe weight $\lambda=10$, the weighted auxiliary-to-prediction loss ratios—used as a proxy for gradient magnitude; we do not measure gradient norms directly—are $15\times$ (collision), $44\times$ (uniform), and $125\times$ (parabola); the encoder’s effective (participation-ratio) dimensionality collapses by 39–90% relative to baseline (uniform $41 \rightarrow 4$). At $\lambda=1$ the ratios are benign, which is the setting used in §4.2.

Checklist. (i) Report per-partition *and* per-horizon; never aggregate across horizons without inspecting them. (ii) Judge with scale-aware prediction metrics (nMSE, decoded-pixel PSNR); use cosine and probe ρ as diagnostics only. (iii) Audit nMSE denominators for degenerate targets. (iv) Calibrate transfer against a random-init control, and validate probe protocols on a random encoder.